

Comparative Metric Learning for Robust Pose-Sequence Retrieval

Updated research-proposal brief

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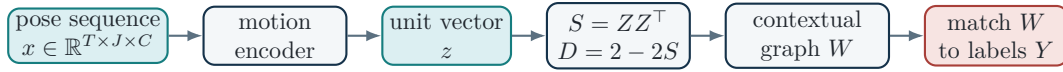
60-second pitch

I will begin with the accepted pipeline: a fixed pretrained motion encoder, a clean reference gallery, and controlled pose corruptions to test whether contextual neighborhoods preserve retrieval better than standard objectives. Once that result is stable, I will keep the encoder and evaluator fixed and map the metric-learning losses evaluated in the 2023 contextual paper into the pose domain. A final screened extension will test selected post-2023 image and pose objectives. The outputs are retrieval and degradation curves plus an applicability map covering performance, robustness, compute, and implementation changes.

Scientific boundary

The foundation paper establishes reduced overfitting and label-noise robustness for *image retrieval*; it does not prove pose-input robustness. Its tables also mix controlled loss re-runs with copied results from architecture-changing methods. The primary pose comparison includes the former, not every row indiscriminately.

Method in one picture



Five equations to narrate

$$\begin{aligned}
 D(i, j) &= \|z_i - z_j\|_2^2 = 2 - 2s_{ij}. \\
 N_{ij} &= \theta(-D(i, j) + \text{sg}(D(i, p_k)) + \epsilon), \quad \frac{\partial \theta}{\partial x} = \alpha. \\
 W_1(i, j) &= \frac{N_{ij}}{2} \left[\frac{M_+(i, j)}{\text{sg} \sum_p N_{ip}} + \frac{M_-(i, j)}{\text{sg} \sum_p (1 - N_{ip})} \right]. \\
 W_2 &= \frac{RW_1}{R\mathbf{1}}, \quad W = \frac{1}{2}(W_2 + W_2^\top), \quad R = N_{k/2} \odot N_{k/2}^\top. \\
 \mathcal{L} &= \lambda \frac{1}{n^2} \sum_{i \neq j} (y_{ij} - w_{ij})^2 + (1 - \lambda) \mathcal{L}_{\text{contrast}} + \gamma \mathcal{L}_{\text{reg}}.
 \end{aligned}$$

The key theoretical statement

For a balanced batch with exactly k samples per class, $n > 2k$, and $\epsilon = 0$, zero contextual loss occurs exactly when every positive ranks ahead of every negative within the batch. This is why k is tied to samples per class rather than treated as an unrelated top- k hyperparameter.

Staged comparison and discussion answers

Controlled experiment

Component	Decision
Dataset	NTU RGB+D 120; official cross-subject and cross-setup protocols.
Encoder	Pretrained MotionBERT sequence encoder; normalized projection output.
Retrieval	Disjoint query/gallery manifest; noisy queries against a fixed clean gallery.
Stage I	Cross-entropy embedding, SupCon, triplet/Multi-Similarity, contextual, contextual+contrastive, and corruption-aware SupCon.
Stage II	Core 2023 reruns: contrastive, mined triplet, Multi-Similarity $\pm$ miner, Proxy Anchor/NCA, ROADMAP, and contextual. Appendix tier: NT-Xent, Fast-/Smooth-AP, then supplemental AP and reported proxy losses.
Stage III	Core recent screen: SGSL, mean-field DML, and PFML. Method-faithful pilots: SimPLE and 2026 NC-Init+PI. Pose controls: continuous-label triplet, MaskCLR, class-aware contrast, and USDRL.
Stage IV	Whole-system transfer, reported separately: IBC, S2SD, Metrix, and HIST first; then DRML, DIML, DiVA, MHGL, and AVSL if resources permit.
Corruptions	Gaussian joint jitter; random joint/limb masking; frame dropout; 3D yaw rotation; pose-estimator shift when available.
Metrics	Recall@1/5/10, mAP, retention $M(c)/M(0)$, and degradation $\Delta(c) = M(0) - M(c)$.
Fairness	Strict track fixes backbone, embedding geometry, sampler, split, evaluator, tuning budget, augmentations, and seeds. Method-faithful variants are reported separately.

Lowest-risk implementation sequence

1. Freeze MotionBERT; validate splits, cosine retrieval, metrics, and deterministic corruptions.
2. Train only small projection heads with SupCon, contextual, and hybrid losses.
3. Complete representative pairwise, proxy, and AP/rank families before appendix-grade losses.
4. Profile time/memory and fine-tune the encoder only after the loss matrix is stable.
5. Select two or three recent methods with distinct mechanisms and reliable implementations.
6. Add pose-specific controls; broaden the dataset only after the NTU study is complete.

Likely questions and precise answers

Question	Answer
What changed after the meeting?	The original contextual robustness study remains Stage I. The new contribution is a controlled transfer map of older and newer optimization families with the same pose encoder.
Does “all 2023 algorithms” mean every table row?	The loss reruns form the causal same-encoder comparison. Copied architecture/framework rows remain a later whole-system transfer tier, with every deviation logged.
Why retrieval rather than classification?	Classification can remain acceptable while the ordering of the embedding space deteriorates. Retrieval measures whether examples remain inspectable and useful for one-shot decisions.
How is applicability measured?	Clean retrieval, degradation under each corruption, time/memory, seed and tuning stability, batch requirements, and whether the method needs changes beyond the loss.
Why two fairness tracks?	Some methods add proxies, queues, learned scorers, masking, or decoders. A strict loss track isolates the objective; a method-faithful track measures practical transfer while logging every deviation.
What would a null or reversed result mean?	It would show that robustness comes mainly from the encoder/corruption exposure, or that image-domain objective rankings do not transfer to temporal pose embeddings.

Do not claim

Do not claim that action-class retrieval measures movement quality, that label-noise robustness implies pose-noise robustness, or that copied state-of-the-art rows are controlled baselines. The contribution is a staged, retrieval-first transfer study of optimization methods.

Closing sentence. *The project begins with one falsifiable robustness question, then asks which metric-learning principles genuinely survive the move from images to pose sequences.*